

# GCSE Mathematics

## Paper 3 (Higher Tier) - Calculator

### Complete Worked Solutions - Questions 1–22

*Every step is shown. Every mark earned.*

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#### Question 1

[3 marks]

**(a) Work out  $1.4^3 + (\sqrt{43.56} - 3.1) \div (1.9 \times 1.7)$**

The expression is  $1.4^3$  PLUS the fraction  $(\sqrt{43.56} - 3.1)$  over  $(1.9 \times 1.7)$ . The  $1.4^3$  is added to the fraction, not inside it.

Step 1:  $1.4^3 = 1.4 \times 1.4 \times 1.4 = 2.744$

Step 2:  $\sqrt{43.56} = 6.6$

Step 3: Numerator of fraction  $= 6.6 - 3.1 = 3.5$

Step 4: Denominator  $= 1.9 \times 1.7 = 3.23$

Step 5: Fraction  $= 3.5 \div 3.23 = 1.083591331...$

Step 6:  $2.744 + 1.083591331... = 3.827591331...$

**Answer: 3.827591331...**

**(b) Round to 3 significant figures**

$3.827591... \rightarrow 3.83$  (3 s.f.)

The 3rd significant figure is 2. The next digit is 7 ( $\geq 5$ ), so round up.

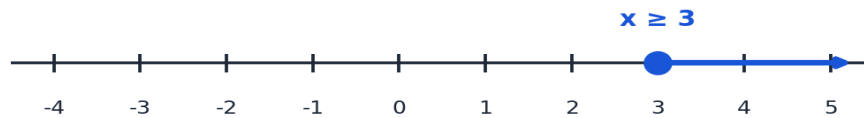
**Answer: 3.83**

**Exam Tip:** Read the layout carefully - is  $1.4^3$  inside or outside the fraction? On a calculator, type:  $1.4^3 + ((\sqrt{43.56} - 3.1) \div (1.9 \times 1.7))$ .

#### Question 2

[3 marks]

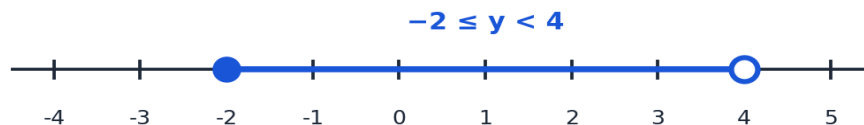
**(a) Write down the inequality**

Question 2(a): Number line showing  $x \geq 3$ 

A filled circle means the value IS included ( $\geq$  or  $\leq$ ). The arrow points right (towards larger values).

**Answer:**  $x \geq 3$

(b) Show  $-2 \leq y < 4$  on a number line

Question 2(b): Number line showing  $-2 \leq y < 4$ 

Filled circle at  $-2$  (included, because  $\leq$ ). Open circle at  $4$  (not included, because  $<$ ). Solid line between them.

**Exam Tip:** Filled = included ( $\leq$  or  $\geq$ ). Open = not included ( $<$  or  $>$ ). This comes up every year.

## Question 3

[4 marks]

(a) Reciprocal of  $0.2$

$$\text{Reciprocal} = 1 \div 0.2 = 5$$

**Answer:**  $5$

(b) Write  $(5^{30})^7$  as a power of  $5$

Power of a power rule: multiply the indices.

$$(5^{30})^7 = 5^{30 \times 7} = 5^{210}$$

**Answer:**  $5^{210}$

(c) Write  $2^{90} \div (2^{10} \times 2^{20})$  as a power of  $2$

$$\text{Denominator: } 2^{10} \times 2^{20} = 2^{30}$$

$$\text{Division: } 2^{90} \div 2^{30} = 2^{60}$$

**Answer:**  $2^{60}$

**Exam Tip:** Multiply  $\rightarrow$  add indices. Divide  $\rightarrow$  subtract. Power of power  $\rightarrow$  multiply.

## Question 4

[3 marks]

50 people did a sponsored walk. Estimate the mean using midpoints.

$$\text{Class } 0-10: \text{ midpoint } 5, f = 8 \rightarrow 5 \times 8 = 40$$

$$\text{Class } 10-20: \text{ midpoint } 15, f = 16 \rightarrow 15 \times 16 = 240$$

$$\text{Class } 20-30: \text{ midpoint } 25, f = 11 \rightarrow 25 \times 11 = 275$$

$$\text{Class } 30-40: \text{ midpoint } 35, f = 8 \rightarrow 35 \times 8 = 280$$

$$\text{Class } 40-50: \text{ midpoint } 45, f = 7 \rightarrow 45 \times 7 = 315$$

$$\text{Total } \Sigma fx = 40 + 240 + 275 + 280 + 315 = 1150$$

$$\text{Mean} = 1150 \div 50 = \pounds 23$$

**Answer: £23**

**Exam Tip:** Always use midpoints. Write your fx column clearly - the examiner wants to see the method.

## Question 5

[4 marks]

Density = Mass  $\div$  Volume, so Mass = Density  $\times$  Volume.

$$\text{Mass of metal bar} = 4 \times 80 = 320 \text{ g}$$

$$\text{Mass of sphere} = 320 \div 2 = 160 \text{ g}$$

$$\text{Volume} = \text{Mass} \div \text{Density} = 160 \div 0.8 = 200 \text{ cm}^3$$

**Answer: 200 cm<sup>3</sup>**

**Exam Tip:** Write the density triangle:  $D = M/V$ . Rearrange for the quantity you need.

## Question 6

[3 marks]

**(a) Change 75 cm<sup>3</sup> to mm<sup>3</sup>**

$$1 \text{ cm} = 10 \text{ mm} \rightarrow 1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$$

$$75 \text{ cm}^3 = 75 \times 1000 = 75\,000 \text{ mm}^3$$

**Answer: 75 000 mm<sup>3</sup>**

**(b) Show 450 km/h = 125 m/s**

$$450 \text{ km} = 450 \times 1000 = 450\,000 \text{ metres}$$

$$1 \text{ hour} = 60 \times 60 = 3600 \text{ seconds}$$

$$\text{Speed} = 450\,000 \div 3600 = 125 \text{ m/s} \quad \checkmark$$

**Answer: 125 m/s (shown)**

**Exam Tip:** For 'show that' questions, write every conversion step.

## Question 7

[3 marks]

20% off means Lethna paid 80% of the original price.

$$80\% \text{ of price} = \pounds 7600$$

$$\text{Price} = 7600 \div 0.80 = \pounds 9500$$

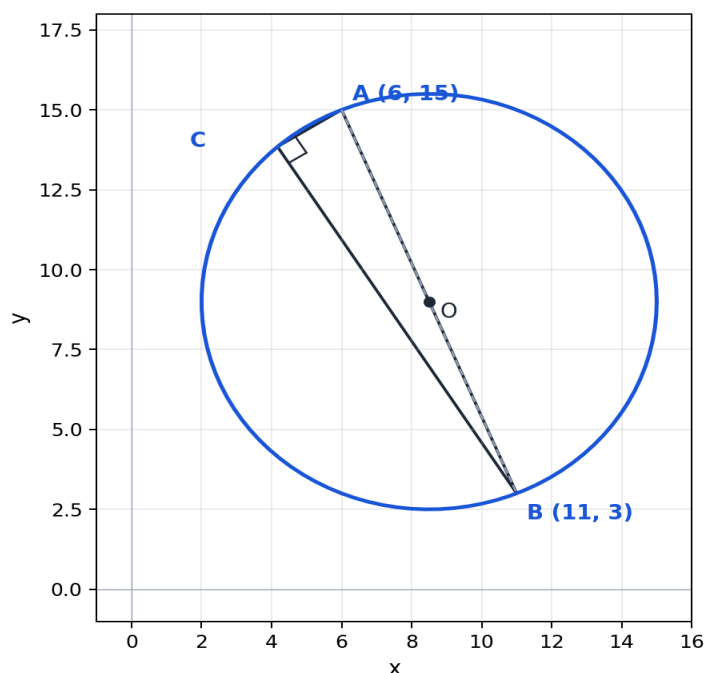
**Answer: £9500**

**Exam Tip:** DON'T find 20% of £7600 and add. The £7600 IS the discounted price.

## Question 8

[4 marks]

Question 8: Circle with diameter AB



AB is the diameter. Find its length using the distance formula.

$$AB = \sqrt{[(11 - 6)^2 + (3 - 15)^2]}$$

$$= \sqrt{[25 + 144]} = \sqrt{169} = 13$$

$$\text{Circumference} = \pi d = 13\pi = 40.840704\dots$$

**Answer: 40.8 (3 s.f.)**

**Exam Tip:** Angle in a semicircle =  $90^\circ$ . The distance formula gives the diameter directly.

## Question 9

[4 marks]

(a) Simplify  $4(x - 2) \div (x - 2)^2$

$$\text{Cancel one } (x - 2): 4(x-2) / (x-2)^2 = 4 / (x - 2)$$

**Answer:  $4 / (x - 2)$**

(b) Make F the subject:  $C = 5(F - 32) / 9$

$$9C = 5(F - 32)$$

$$9C/5 = F - 32$$

$$F = 9C/5 + 32$$

**Answer:  $F = 9C/5 + 32$**

## Question 10

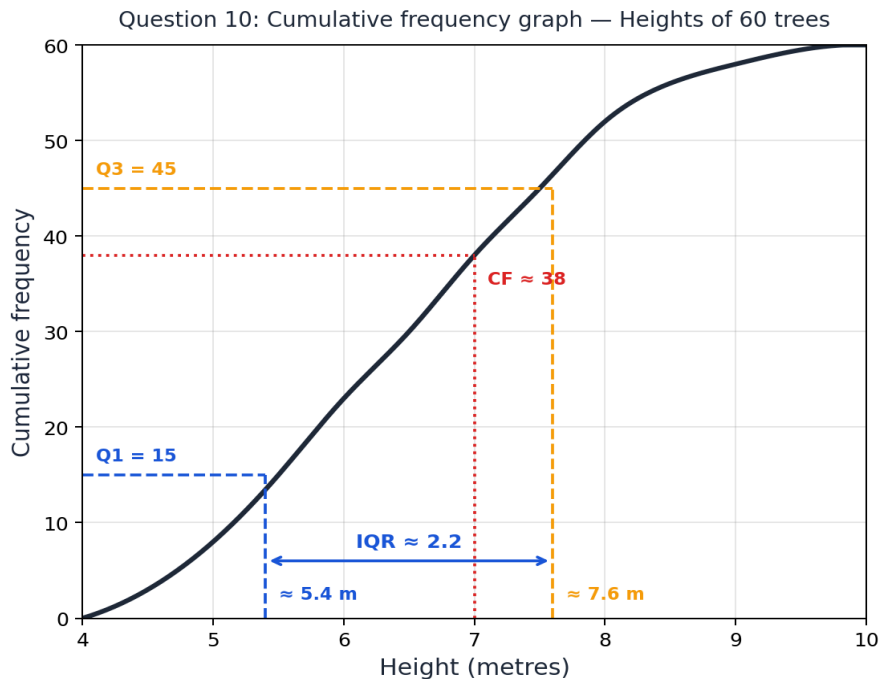
[6 marks]

(a) Is Chris correct?

Chris says " $10 + 40 = 50$ " but the values are already CUMULATIVE. The 40 already includes the 10.

**Answer:** No. The cumulative frequency for speed  $\leq 90$  is already 40, not  $10 + 40 = 50$ .

**(b) Interquartile range from the cumulative frequency graph**



**60 trees total. Find Q1 and Q3 positions:**

Q1 position =  $60 \div 4 = 15$ th value

Q3 position =  $3 \times 60 \div 4 = 45$ th value

Reading from the graph (see diagram above):

At CF = 15: height  $\approx 5.4$  m (Q1)

At CF = 45: height  $\approx 7.6$  m (Q3)

IQR = Q3 - Q1 =  $7.6 - 5.4 = 2.2$  m

**Answer:** IQR  $\approx 2.2$  m (accept 1.8 to 2.6)

**(c) Trees with height greater than 7 m**

At height = 7 m, CF  $\approx 38$  (see red dashed line on graph)

Trees  $> 7$  m =  $60 - 38 = 22$

**Answer:**  $\approx 22$  trees (accept 18 to 24)

**Exam Tip:** Draw your reading lines on the graph with a ruler - the examiner needs to see HOW you got your values.

## Question 11

[3 marks]

$W = m^2/p$ . Use multipliers for percentage changes.

New  $m = 0.6m$  (decreased by 40%)

New  $p = 1.5p$  (increased by 50%)

New  $W = (0.6m)^2 / (1.5p) = 0.36m^2 / 1.5p = 0.24W$

Decrease =  $1 - 0.24 = 0.76 = 76\%$

**Answer:** 76%

**Exam Tip:** Multipliers (0.6, 1.5) are much cleaner than picking random numbers.

## Question 12

[6 marks]

(a)  $h$  inversely proportional to  $g$ :  $hg = k$

$$k = 2 \times 150 = 300$$

$$\text{When } g = 5: h = 300/5 = 60$$

$$\text{When } h = 10: g = 300/10 = 30$$

**Answer:**  $h = 60$  when  $g = 5$  |  $g = 30$  when  $h = 10$

(b) Show  $p = 108$  when  $r = 2n$

$$p = kr^3. \text{ When } r = n: 13.5 = kn^3 \rightarrow k = 13.5/n^3$$

$$\text{When } r = 2n: p = (13.5/n^3)(2n)^3 = (13.5/n^3)(8n^3) = 108 \checkmark$$

**Answer:**  $p = 108$  (shown)

**Exam Tip:**  $(2n)^3 = 8n^3$ . Doubling the input multiplies the cube by 8.

## Question 13

[3 marks]

Rearrange  $x^2 - 3x - 2 = 0$  to use the graph  $y = x^2 - 4x$ .

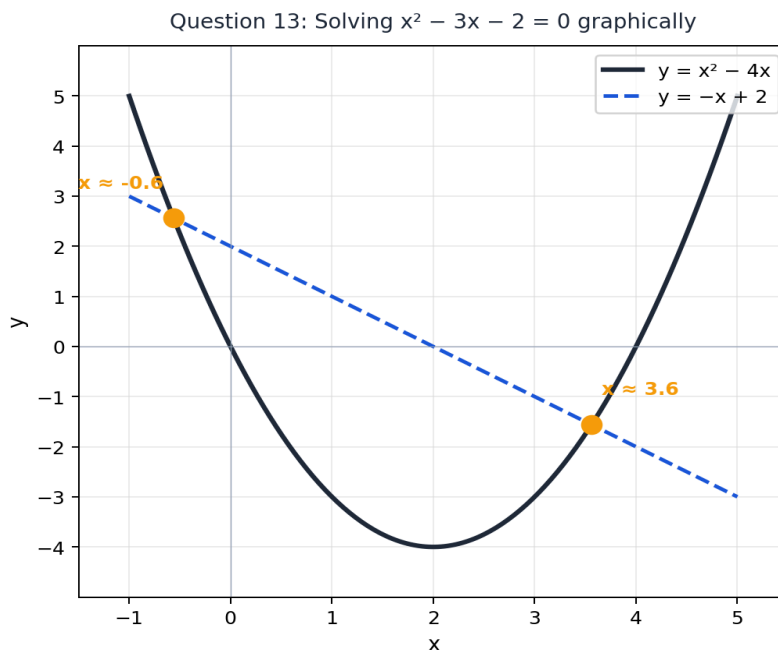
$$x^2 - 3x - 2 = 0$$

$$(x^2 - 4x) + x - 2 = 0$$

$$y + x - 2 = 0$$

$$y = -x + 2 \leftarrow \text{draw this line}$$

The line  $y = -x + 2$  passes through  $(0, 2)$  and  $(2, 0)$ . Draw it on the graph and read the intersections.



The orange dots show where the line crosses the parabola:

**Answer:**  $x \approx -0.6$  and  $x \approx 3.6$  (accept  $\pm 0.1$ )

**Exam Tip:** Show how you derived the line. That derivation IS the method mark.

## Question 14

[3 marks]

Express  $1/\sqrt{20}$  as  $2^x \times 5^y$ .

$$\begin{aligned} 1/\sqrt{20} &= 20^{-1/2} = (2^2 \times 5)^{-1/2} \\ &= 2^{-1} \times 5^{-1/2} \end{aligned}$$

**Answer:**  $x = -1, y = -1/2$

**Exam Tip:** Break 20 into prime factors first ( $2^2 \times 5$ ), then apply the power to each.

## Question 15

[3 marks]

Let  $b = 4n$ . Then  $a = 4n - 4$ ,  $c = 4n + 4$ .

$$b^2 = (4n)^2 = 16n^2$$

$$ac = (4n-4)(4n+4) = 16n^2 - 16 \text{ (difference of two squares)}$$

$$b^2 - ac = 16n^2 - (16n^2 - 16) = 16$$

Therefore  $ac$  is always exactly 16 less than  $b^2$ . ■

**Exam Tip:** Spot the difference of two squares:  $(b-4)(b+4) = b^2 - 16$ .

## Question 16

[5 marks]

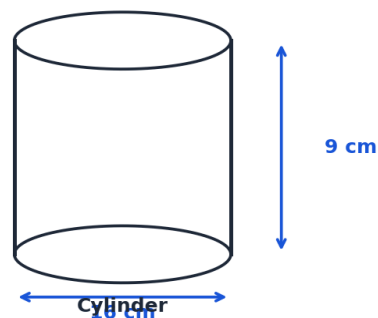
Question 16: Hemisphere and Cylinder

**Volume ratio = 1 : 4**



$r = ?$

Hemisphere



Cylinder

**Cylinder: radius = 8 cm, height = 9 cm.**

$$\text{Vol. cylinder} = \pi(8)^2(9) = 576\pi$$

$$\text{Vol. hemisphere} = 576\pi \div 4 = 144\pi \text{ (ratio 1:4)}$$

$$(2/3)\pi r^3 = 144\pi \rightarrow r^3 = 216 \rightarrow r = 6 \text{ cm}$$

$$\text{Total surface area} = \text{curved} + \text{flat base} = 2\pi r^2 + \pi r^2 = 3\pi r^2$$

$$= 3\pi(36) = 108\pi$$

**Answer:**  $108\pi \text{ cm}^2$

**Exam Tip:** Don't forget the flat base of the hemisphere! Total SA =  $2\pi r^2 + \pi r^2 = 3\pi r^2$ .

## Question 17

[4 marks]

$E = (a - c)/m$ . Find the lower bound.

$$a = 35 \text{ (2 s.f.): } 34.5 \leq a < 35.5$$

$$c = 2.3 \text{ (2 s.f.): } 2.25 \leq c < 2.35$$

$$m = 12.6 \text{ (3 s.f.): } 12.55 \leq m < 12.65$$

**Lower bound of E: minimise numerator, maximise denominator.**

$$\text{Min}(a - c) = 34.5 - 2.35 = 32.15$$

$$\text{Max}(m) = 12.65$$

$$\text{Lower bound} = 32.15 \div 12.65 = 2.54150...$$

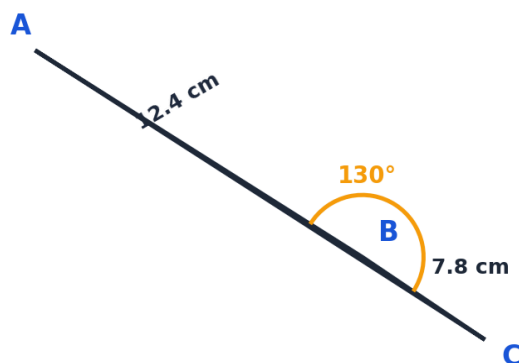
**Answer: 2.54 (3 s.f.)**

**Exam Tip:** To minimise a fraction: smallest top, largest bottom.

## Question 18

[3 marks]

Question 18: Triangle ABC



Area =  $\frac{1}{2}ab \sin C$  (angle between the two known sides).

$$= \frac{1}{2} \times 12.4 \times 7.8 \times \sin 130^\circ$$

$$= 48.36 \times 0.76604...$$

$$= 37.04...$$

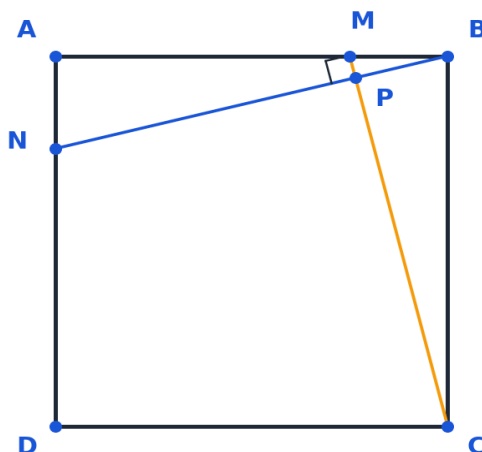
**Answer: 37.0 cm<sup>2</sup> (3 s.f.)**

**Exam Tip:**  $\sin 130^\circ = \sin 50^\circ \approx 0.766$ . Obtuse angles work the same way in the formula.

## Question 19

[5 marks]



Question 19: Square ABCD with  $BN \perp CM$ 

**Prove that  $BM = AN$ .**

Step 1: Angle relationships from perpendicularity at P.

In  $\triangle BPC$ :  $\angle BPC = 90^\circ$

$\therefore \angle NBC + \angle BCM = 90^\circ \dots (i)$

Step 2: Use the right angle at B (corner of the square).

$\angle ABC = 90^\circ \rightarrow \angle ABN + \angle NBC = 90^\circ \dots (ii)$

From (i) and (ii):  $\angle ABN = \angle BCM$

**Step 3: Prove congruent triangles.**

In  $\triangle ABN$  and  $\triangle BCM$ :

$\angle NAB = \angle MBC = 90^\circ$  (corners of square)

$AB = BC$  (sides of square)

$\angle ABN = \angle BCM$  (proved above)

By ASA:  $\triangle ABN \cong \triangle BCM$

**Therefore:  $AN = BM$  ■**

**Exam Tip:** For proofs: state fact  $\rightarrow$  give reason  $\rightarrow$  link to conclusion. Every line needs justification.

## Question 20

[5 marks]

**Step 1: Find intersection A.**

L:  $y = 2x - 2$  into M:  $2x + 3(2x - 2) = 18$

$8x - 6 = 18 \rightarrow x = 3, y = 4$ . Point A = (3, 4)

**Step 2: Perpendicular gradient.**

Gradient of M =  $-2/3 \rightarrow$  Perpendicular =  $3/2$

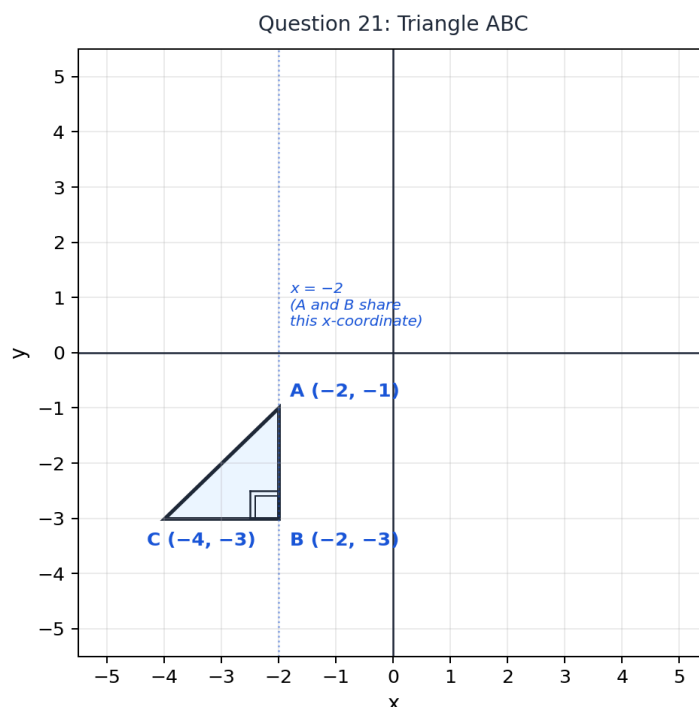
**Step 3: Equation through A(3, 4).**

$y - 4 = (3/2)(x - 3) \rightarrow y = (3/2)x - 1/2$

**Answer:  $y = (3/2)x - 1/2$**

## Question 21

[4 marks]



A and B are invariant (unchanged) after both transformations. C is not.

**The crucial observation is that A and B share the same x-coordinate; they lie on a vertical line.**

Step 1: The translation by  $(k, 0)$  shifts every point  $k$  units horizontally.

Step 2: Since A and B share the same x-coordinate, they both shift by exactly the same amount. A vertical line of reflection placed halfway between their original and translated positions will map both back to their starting points.

Step 3: C does NOT share that x-coordinate, so it reflects to a different position. C is not invariant.

✓

**Answer:** Reflection in the line  $x = k/2 - 2$  (where  $-2$  is the shared x-coordinate of A and B)

**Exam Tip:** When two points are invariant under a combined transformation, look for what they have in common - here, the same x-coordinate.

## Question 22

[4 marks]

5 red,  $y$  yellow sweets.  $P(\text{two yellow}) = 3/7$ . Without replacement.

$$y/(5+y) \times (y-1)/(4+y) = 3/7$$

$$7y(y-1) = 3(5+y)(4+y)$$

$$7y^2 - 7y = 60 + 27y + 3y^2$$

$$4y^2 - 34y - 60 = 0 \rightarrow 2y^2 - 17y - 30 = 0$$

$$(2y+3)(y-10) = 0 \rightarrow y = 10$$

Total = 15 sweets.

$$P(\text{two red}) = 5/15 \times 4/14 = 20/210 = 2/21$$

**Answer:** 2/21

**Exam Tip:** Form the quadratic from the probability tree. Don't forget 'without replacement' changes the denominator.

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